6.3 The Powerhouse of the Cell

Mitochondria are the sites of cellular respiration, the process that uses oxygen to break down sugar and release the energy stored in food as ATP. A single cell may contain hundreds or even thousands of mitochondria, depending on how much energy it needs.

Each mitochondrion is enclosed by two membranes. The inner membrane is folded into cristae, which greatly increase the surface area available for respiration. Mitochondria also carry their own small, circular DNA, a clue to their ancient origin as free-living bacteria.

Muscle and nerve cells, which use large amounts of energy, are especially rich in mitochondria. Plant cells contain mitochondria as well as chloroplasts, which capture light energy for photosynthesis.

Key terms
cellular respiration · ATP · cristae · mitochondrial DNA

Check yourself
1. Why are muscle cells rich in mitochondria?
2. What is the function of the cristae?